

TrES-3b

R-2029

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130417 · created 2026-08-08T01:00:49+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456399.8971 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.013
Transit depth (Rp/Rs)^2	2.84 +/- 0.44 [%]
Orbital Inclination (inc)	83.36 +/- 0.65
Airmass coefficient 1 (a1)	0.993 +/- 0.013
Airmass coefficient 2 (a2)	0.006 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#2 - [547, 114]
Optimal Aperture	3.66
Optimal Annulus	10.24
Transit Duration (day)	0.0668 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.013 vs 0.16309 (deviation 0.45 σ)
Duration fit vs published	0.0668 d vs 0.0567 d (deviation 2.3 σ)
Mid-time O-C	-0.4 min
Depth SNR	13.0
Residual scatter	1.35 %

Light curve

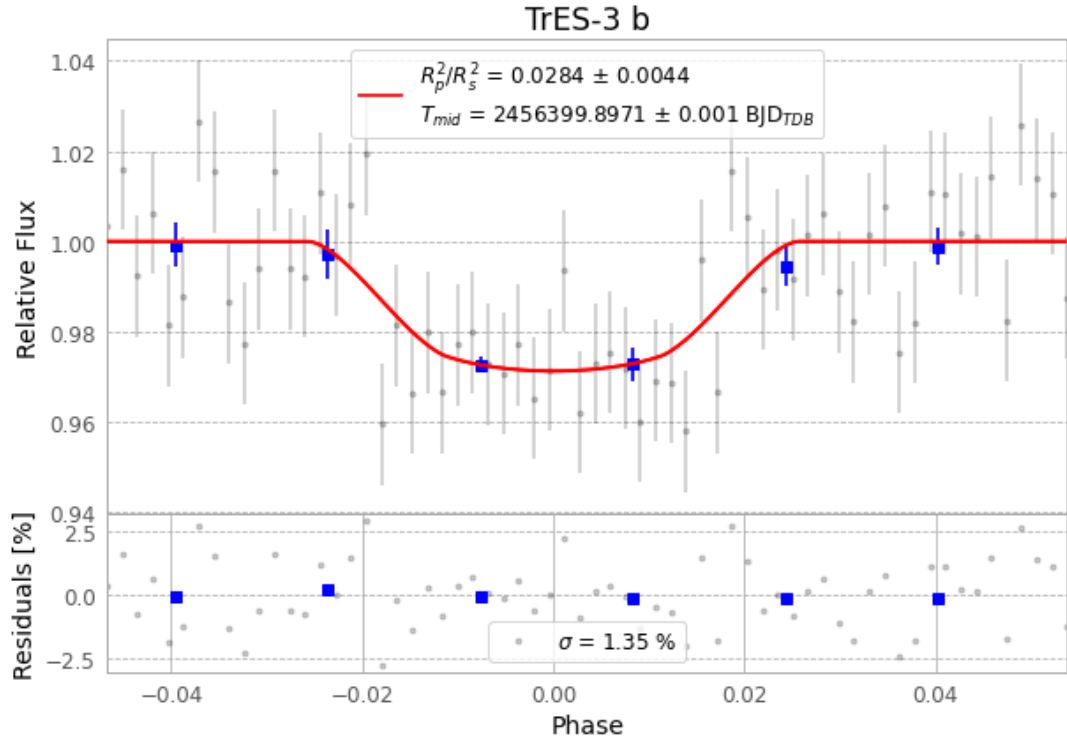


Figure 1 — FinalLightCurve_TrES-3 b_2013-04-17.png (SHA-256 905df3b8a676b3a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f3c02b09e4d4251...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), TRES-3130417080012.FITS → TRES-3130417110913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130417.log (1732417c6d23...), FinalLightCurve_TrES-3 b_2013-04-17.png (905df3b8a676...), FinalLightCurve_TrES-3 b_2013-04-17.csv (fb094d7802f2...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:00Z.